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Reading Your Actions: Learning Generalizable Action Representations via Pre-training AEMG

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Abstract

A fundamental role in decoding human motor intent and enabling intuitive human–computer interaction is played by electromyography (EMG). However, its generalization capability across subjects, devices, and tasks remains substantially limited by data heterogeneity, label scarcity, and the lack of a unified representational framework. To bridge this gap, we propose Any Electromyography (AEMG), the first large-scale, self-supervised representation learning framework for EMG. AEMG reconceptualizes neuromuscular dynamics linguistically, utilizing a novel Neuromuscular Contraction Tokenizer (NCT) to translate discrete muscle contractions into structural words and temporal activation patterns into coherent sentences. Furthermore, we compile the largest cross-device EMG signal vocabulary to date, enabling seamless transfer across arbitrary channel topologies and sampling rates. Experiments demonstrate that AEMG improves the zero-shot leave-one-subject-out (LOSO) accuracy by 5.79–9.25% *compared to six state-of-the-art baselines, and achieves more than* 90% *few-shot adaptation performance with only* 5% *of target user data. Our work has proposed the concept of EMG signals as a cross-device physiological language, learned their grammar from massive amounts of data, and laid the groundwork for a single-training, universally applicable EMG foundation model. Our code, demos, and more resources are available at* https://github.com/AEMG-series/AEMG.

1. Introduction

The spontaneous electrical activity of muscles is recorded by electromyography (EMG), a non-invasive technique that serves as a crucial signal source for constructing myoelectric human-computer interfaces and understanding motor control. Common non-invasive surface EMG (sEMG) signals are acquired using dry or wet electrode arrays attached to the skin surface. This signal can be abstracted as a real-valued matrix $I \in \mathbb{R}^{C\times T}$, where C represents the number of electrodes (channel), which varies with the device configuration, and T denotes the number of temporal sampling points, determined by the sampling rate and trial duration [1, 21]. As a highly objective physiological signal, EMG demonstrates significant potential across various human-computer interaction (HCI) domains. These include advanced prosthetic control, assistive robot teleoperation, gesture-based diagnosis of neuromuscular diseases, and motor intent decoding for individuals with paralysis or amputation [1, 4, 20, 24].

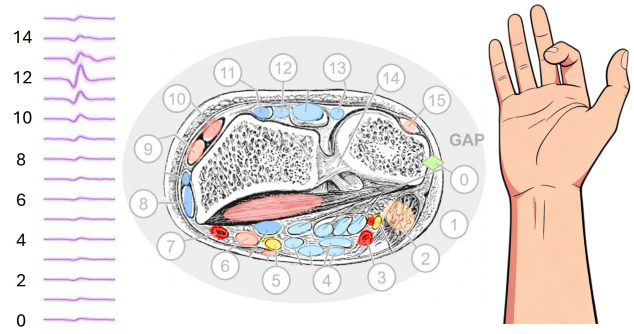


Figure 1. Corresponding electromyography, gesture, and anatomical diagram during index finger flexion

In current research, numerous deep learning models have been proposed [2, 27]. However, despite the excellent performance of those on specific myoelectric tasks, these models primarily focus on EMG samples adhering to particular task formats. Their generalizability remains constrained by multiple challenges at the data, device, and paradigm levels [1]. Current research lacks a universal self-supervised paradigm for heterogeneous EMG, confining existing models to specific tasks and small-scale data. This ultimately results in a specialized rather than general-purpose limitation, making it difficult to achieve cross-task learning and rapid adaptation for new users [20].

Recently, we have been significantly inspired by the ca-

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pabilities of Large Language Models (LLMs) [19, 25, 28]. Specifically, Transformer-based models have achieved remarkable results in natural language processing (NLP) tasks, highlighting the potential of self-supervised pre-training as an effective means to leverage large-scale data. Motivated by these methods, our study attempts to apply the reconstruction concept to pre-train a neural transformer for learning general-purpose myoelectric representations. However, directly applying LLM-style pre-training to EMG data is a daunting task.

The challenges are summarized as three points.Publicly available EMG datasets exhibit substantial variations in acquisition hardware (e.g., electrode types, channel counts, topological layouts) and signal characteristics (e.g., sampling rates, recording protocols). Secondly, the intrinsic non-stationarity of EMG signals severely limits cross-subject and cross-session generalization, exposing a critical void in robust representation learning paradigms. Additionally, parsing raw EMG streams into physiologically grounded and semantically coherent primitives remains a fundamental bottleneck. Standardizing such discrete representations across subjects and devices is essential for building generalizable myoelectric models [11, 29].

Our work aims to develop a foundational model for general myoelectric understanding through large-scale self-supervised pre-training on multi-source EMG data.The main contributions are summarized as follow:

- **Bridging hardware heterogeneity via a unified representation space.** We propose a rigorous data-rectification pipeline that projects 8 highly heterogeneous EMG datasets into a standardized input space, explicitly aligning varying channel counts and diverse sensor topologies. To our knowledge, this marks the first time such a massive and diverse collection of unannotated datasets has been aggregated for large-scale EMG pre-training.
- **A linguistic paradigm for non-tationary motor intentions.** Breaking away from traditional sliding-window heuristics, we introduce a novel linguistic perspective tailored to voluntary muscle contractions. By conceptualizing asynchronous muscle activation bursts as semantic words and their synergistic multi-channel coordination as sentences, we construct the largest cross-subject EMG vocabulary to date. This paradigm naturally resolves the non-stationary characteristics of sEMG and intrinsically links surface morphologies to complex gesture syntax
- **AEMG: A synergy-aware self-supervised framework.** Rather than relying on periodic templates, we first discretize dynamic neuromuscular primitives into a compact, semantically meaningful codebook. Subsequently, through a masked reconstruction objective, the framework explicitly models the spatiotemporal dependencies and underlying muscle synergies across varying electrode locations.

Building upon the aforementioned design, this work aims to address three key questions:

- **Q1:** How can large-scale, heterogeneous, and unlabeled EMG data be leveraged to learn generalizable representations that are compatible across devices and tasks?
- **Q2:** Given the high heterogeneity in channels and topologies, what neural tokenization-masking paradigm best preserves the spatio-temporal semantics of EMG?
- **Q3:** To what extent can unified pre-training improve performance and data efficiency for essential downstream tasks such as gesture recognition under cross-subject and cross-session settings?

2. Related Work

Deep learning has achieved significant breakthroughs in myoelectric decoding accuracy. However, the model's generalization capability and rapid adaptability to new users remain the core bottlenecks hindering large-scale practical application. Existing research can be broadly divided into three evolutionary stages.

The Individualized Modeling and Intra-Dataset Optimization Stage. Early machine learning and deep learning efforts focused on enhancing recognition accuracy within individual datasets. These studies typically employed random-split or within-subject-split evaluation methods [1, 16]. Although reported accuracies were high, often exceeding 95%, these results largely failed to reveal the models' true performance when encountering new users [40]. This high accuracy was often inflated. Simultaneously, these models overfit to irrelevant variables. This resulted in a drastic performance decline under strict leave-one-subject-out cross-validation (LOSO-CV), where accuracies commonly fell below 50% [1, 2, 6, 10, 16, 39, 40]. The work from this stage advanced the evolution of model architectures, but it essentially remained within the paradigm of customizing models for each specific scenario.

Models for Cross-Domain Generalization and Few-Shot Adaptation. To address the generalization challenges identified in the first stage, researchers began designing algorithms and models specifically to enhance performance on new users or new sessions under zero-shot and few-shot scenarios [34]. Research during this phase primarily employed Unsupervised Domain Adaptation (UDA) methods.

UDA methods can be further categorized into conservative and non-conservative strategies. Conservative approaches retain source domain data during adaptation and focus on aligning the distributions of the source and target domains [5, 7, 13, 23]. For instance, some studies utilized deep adversarial networks to learn domain-invariant features, thereby achieving marginal distribution alignment [15, 37]. However, the limitations of conservative strategies become apparent when significant concept drift exists in cross-subject EMG signals. Non-conservative methods

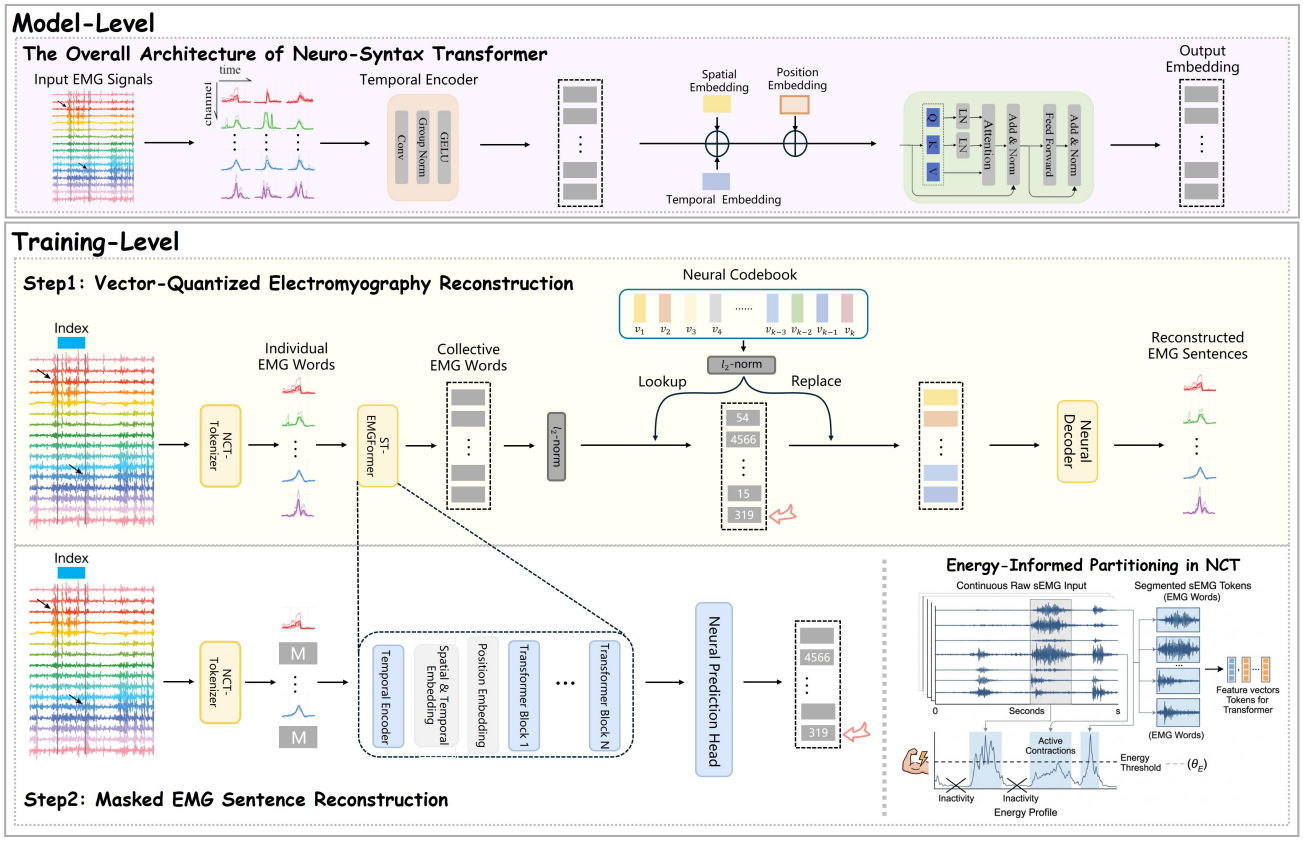


Figure 2. Framework of AEMG

overcome this limitation by focusing entirely on the target domain samples [9, 31].

The Exploration Stage of General-Purpose Model Paradigms Driven by Ultra-Large-Scale Data. It is pioneered by leading teams such as Meta's CTRL-labs. The core idea borrows from the concept of foundation models in the field of Natural Language Processing: a model's powerful generalization capability ultimately stems from pre-training on massive and diverse data [20].

The field of EMG decoding is currently at a critical juncture, undergoing a paradigm shift. Although the dawn of general-purpose models has emerged, a significant research gap persists: How can massive volumes of unlabeled or weakly-labeled [20], heterogeneous myoelectric data be leveraged. Our study aims to fill this gap. By integrating multiple heterogeneous public datasets, a lower-cost and more scalable path toward building an EMG foundation model is explored.

3. Method

This chapter elaborates on the construction of the AEMG framework, a self-supervised pre-training paradigm for multi-source heterogeneous EMG data. The framework is designed to learn general spatiotemporal joint EMG representations by leveraging large-scale unlabeled signals. The AEMG construction process encompasses five integral components from data processing to the core pre-training stage, as illustrated in Figure 2.

3.1. Data Collection and Preprocessing

Eight publicly available EMG datasets focused on gesture recognition are extensively collected, encompassing diverse electrode types, subject populations, and acquisition protocols [6, 12, 18, 22, 26]. The original datasets exhibit substantial variation in dominant frequency bands, with spectral properties influenced by factors including skin-electrode impedance, muscle fiber type composition, and muscle recruitment patterns. Spatial characteristics reflect selective activation of distinct muscle groups, such as forearm flexors and extensors. These spatiotemporal features collectively encode gesture and movement information.

The acquisition setup involves widely adopted devices, forming a highly heterogeneous ecosystem in terms of channel count, electrode material, sampling rate, and wearability. Sensor configurations, corresponding specifications, and subject counts per dataset are summarized in Appendix Figure 1. Further dataset details are provided in Ap-

pendix B.1 and Appendix Tables 2. The aggregated dataset comprises a large volume of unlabeled EMG data from over 500 person-time, covering diverse electrode configurations and a broad spectrum of gesture categories.

3.2. Neural Contraction Tokenizer

Significant heterogeneity exists across EMG datasets in terms of channel count, sampling rate, gesture categories, and recording hardware, which substantially hinders unified modeling. To address this, a unified preprocessing pipeline is designed to project all EMG data into a consistent input space, thereby enabling large-scale cross-task pre-training.

Inspired by QRS-based tokenization approaches [19], NCT is introduced to convert raw EMG signals into muscle tokens. The detailed procedure of the energy-based segmentation and tokenization is summarized in Algorithm 1.

Given a raw signal $\mathbf{X}_i \in \mathbb{R}^{C_i \times T_i}$, local muscle contraction activity is first detected. A sliding window energy threshold is applied to determine the presence of contraction:

$$E_w = \frac{1}{L_w} \sum_{t=1}^{L_w} \sum_{c=1}^{C_i} X_i(c,t)^2 \tag{1}$$

A segment is labeled as a valid muscle contraction if $E_w > \theta$. The threshold θ is adaptively set based on the resting state noise level of each specific subject during a brief calibration period. Each valid segment is defined as a neural token, denoted as:

$$\mathbf{U}^{(k)} \in \mathbb{R}^{C_i \times L_k}, \quad k = 1, \ldots, N \tag{2}$$

Within-Segment Normalization To mitigate inter-subject and inter-channel variability, each segment undergoes z-score normalization:

$$\hat{U}^{(k)}(c,t) = \frac{U^{(k)}(c,t) - \mu^{(k)}(c)}{\sigma^{(k)}(c) + \epsilon} \tag{3}$$

Channel Mapping and Interpolation To handle inconsistent channel arrangements across datasets, a fixed channel mapping function $\mathcal{R}(\cdot)$ is introduced, which rearranges channels into a predefined topological order.

3.3. NST Backbone Network

The NCT is pivotal for converting continuous raw EMG signals into semantic EMG sentences. To effectively capture spatiotemporal features and latent semantic relationships within EMG sentences, a backbone network named Neuro-Syntax Transformer(NST) is designed. This backbone is employed across all core components of the AEMG framework, including neural vocabulary construction (Section 3.4), masked modeling pre-training (Section 3.5), and fine-tuning for downstream tasks.

Algorithm 1 Neural Contraction Tokenizer

Require: Raw EMG signal $X \in \mathbb{R}^{C \times T}$, Window size L_w, Threshold θ, Max Length l_{max}
Ensure: EMG Sentence $I \in \mathbb{R}^{l_{max} \times d}$

```
Initialize token list 𝒰 ← []
for t = 1 to T with step stride do
    Extract window w_t = X[:, t : t + L_w]
    Calculate energy E_w = (1/L_w) Σ_{c,τ} (w_t[c, τ])^2
    if E_w > θ then
        Identify Contraction: Mark segment as valid token U^(k)
        Normalize: U^(k) ← z-score(U^(k))
        Project: u^(k) ← 1D-Conv(U^(k))
        Append u^(k) to 𝒰
    end if
end for
Construct Sentence: I ← Concat(𝒰)
if len(I) < l_max then
    Pad I with zero tokens to length l_max
else
    Truncate I to length l_max
end if
return I
```

Due to the transient, broadband nature of sEMG, each neuromuscular token encodes detailed kinetic information. Denoting the raw multi-channel token sequence as $\mathbf{X}$, we project those into a latent semantic space via a 1-D convolutional layer, moving beyond raw amplitude dependence.

$$\mathbf{I}_t = \text{GELU}(\mathbf{W}_{\text{conv}} * \mathbf{X} + \mathbf{b}_{\text{conv}}) \tag{4}$$

where $*$ denotes the convolution operator. This mapping yields a sequence of continuous feature embeddings, denoted by $\mathbf{I}_t \in \mathbb{R}^{l \times d}$ [8].

A crucial innovation of our architecture is the explicit modeling of muscle synergies across highly heterogeneous sensor topologies. To achieve this, we dynamically enrich each contraction token with a joint spatiotemporal condition space. Specifically, this space explicitly encodes three critical attributes: the anatomical source (electrode layout), the precise activation phase, and the chronological order [19]. Finally, these structural embeddings are linearly fused to yield a unified, context-aware semantic representation $\mathbf{I}'$.

Subsequently, $\mathbf{I}'$ is processed by a Transformer encoder via self-attention to capture long-range neuromuscular correlations [17, 36]:

$$\{Q, K, V\} = \{\text{LN}(\mathbf{I}')W^Q, \text{LN}(\mathbf{I}')W^K, \text{LN}(\mathbf{I}')W^V\}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_{\text{head}}}}\right)V \tag{5}$$

3.4. Neural Codebook Construction and Reconstruction Training via Vector Quantization

The foundational contraction units identified by the NCT (Section 3.2) are intrinsically biased by user-specific physiology and acquisition conditions, precluding broad generalization. We introduce a universal myoelectric vocabulary—a discrete neural codebook of canonical muscle activations. Forcing the model to map diverse input signals to these shared physiological archetypes enables the AEMG framework to extract highly generalizable, cross-subject representations during pre-training.

While surface electromyograms are highly individualized, the neuromuscular recruitment strategies for identical motor tasks share a fundamental underlying topology [20]. We construct our cross-domain myoelectric vocabulary by jointly optimizing vector quantization and signal reconstruction [17, 19].

3.4.1. Vector Quantization

We parameterize the discrete token space through an EMG codebook $\mathcal{V} = \{\mathbf{v}_i\}_{i=1}^{k} \in \mathbb{R}^{k\times d}$, comprising k collective word representations of size d. In the forward pass, an initial EMG segment $\mathbf{X}_i \in \mathbb{R}^{C_i\times T_i}$ undergoes tokenization via NCT [19], resulting in a symbolic sentence $\mathbf{I} \in \mathbb{R}^{l\times t}$. Driven by the NST architecture (Section 3.3), this sentence is further encoded into a set of continuous hidden states $\mathbf{P} = \{\mathbf{p}_i \mid i = 1, \ldots, l\}$.

A quantizer is subsequently employed to discretize these continuous vectors $\mathbf{p}_i$ into collective token embeddings. The EMG vocabulary $\mathcal{V}$ is searched to find the nearest neighbor for each $\mathbf{p}_i$ using the L2-norm distance metric.

$$z_i = \| \, \mathrm{l}_2(\mathbf{p}_i) - \mathrm{l}_2(\mathbf{v}_j)\|_2^2 \tag{6}$$

where $\mathbf{v}_j$ denotes the j-th collective token embedding in the vocabulary, z_i represents the quantized vocabulary index for $\mathbf{p}_i$.

3.4.2. Neural Contraction Reconstruction

After quantization assigns index labels z_i, the normalized discrete collective token embeddings $\{l_2(\mathbf{v}_{z_i}) \mid i = 1, \ldots, l\}$ are passed to a transformer decoder f_d. This reconstruction process is expressed as:

$$\hat{\mathbf{I}} = \bigcup_{i=1}^{l} f_d\left(l_2(\mathbf{v}_{z_i})\right) \tag{7}$$

where $\hat{\mathbf{I}}$ denotes the reconstructed EMG sentence, and the union symbol $\cup$ indicates the sequential decoding and composition performed by decoder f_d over the sequence $i = 1, \ldots, l$.

To stabilize the EMG vocabulary updates, an exponential moving average strategy is employed for optimization. The quantization and reconstruction processes are guided by a mean squared error loss [17]. The overall loss function $\mathcal{L}_{VQ}$ for training the vector quantization process is defined as:

$$\mathcal{L}_{VQ} = \sum_{l\in\mathcal{D}}\sum_{i=1}^{l}\Bigg(\underbrace{\|\hat{I}_i - I_i\|_2^2}_{\text{(1) Reconstruction Loss}} + \underbrace{\|\mathrm{sg}(l_2(p_i)) - l_2(v_{z_i})\|_2^2}_{\text{(2) Vocabulary Learning Loss}} + \underbrace{\|l_2(p_i) - \mathrm{sg}(l_2(v_{z_i}))\|_2^2}_{\text{(3) Encoder Commitment Loss}}\Bigg) \tag{8}$$

where $\mathcal{D}$ represents the complete set of EMG training samples, $\mathbf{I}_i$ denotes the i-th token in the original EMG sentence, $\hat{\mathbf{I}}_i$ represents its reconstructed counterpart, and sg indicates the stop-gradient operation that acts as an identity function during forward propagation while having zero gradient during backpropagation.

3.5. Cross-patch Masked Modeling for Pre-training

The pre-training process of the AEMG framework is illustrated in Step 2 of the lower section of Figure 2. During this stage, the framework learns generalized spatiotemporal-level and synergistic-level muscle activation representations by masking a subset of individual muscle contraction tokens in EMG sentences and predicting the corresponding collective muscle contraction tokens at the masked positions based on the unmasked context. More detailed mask training derivation and formulas are provided in Appendix D.6 and its subsections.

4. Experiments

To systematically evaluate the effectiveness of the AEMG framework, the experimental setup follows the benchmark configurations and results provided by Yang *et al.* [40], adopting a LOSO-CV paradigm. The data from each subject is iteratively used as the target domain, while the data from all other subjects are aggregated as the source domain. This setup constitutes the most stringent standard for examining the model's cross-subject generalization capability.

4.1. Zero-shot Generalization: Leave-One-Subject-Out Performance

Table 1 presents the LOSO-CV results of AEMG against six other SOTA methods on the four benchmark gesture classification datasets. Under the LOSO setting, the model is tested on the entire data of one subject and trained on the data from all other subjects.

On all evaluated datasets, the AEMG framework largely outperforms other SOTA models. Specifically, while performance varies across datasets—for instance, MDD achieves the best SOTA performance on Dataset 2. But, our AEMG-Large model achieves a significant improvement in average accuracy across all datasets.

Table 1. Comparison of LOSO accuracy (%) between AEMG and other methods on gesture classification tasks.

Method	Dataset 1 [32] (ULB-MLG)	Dataset 2 [3] (EMG-EPN)	Dataset 3[30] (Ninapro DB4)	Dataset 4 [35] (Toro-Ossaba)	Average
Intra-Subject	93.36	98.11	90.50	88.82	92.70
Cross-Subject	62.35	77.06	48.50	82.05	67.49
Normalization	80.33	84.73	75.83	86.93	81.96
MDD[41]	82.67	**88.97**	64.67	87.13	80.86
CDEM[13]	83.00	86.95	81.33	84.55	83.96
SCDEM[38]	82.82	86.75	82.33	84.17	84.02
VADA[31]	78.04	88.68	72.83	84.48	81.01
VADA + DIRT-T[31]	78.12	88.57	70.50	85.06	80.56
AEMG-Base (Ours)	88.52	87.10	81.21	89.13	86.49
AEMG-Large (Ours)	**91.50**	88.32	**88.10**	**91.30**	**89.81**

Table 2. Few-shot adaptation of AEMG-Large with varying fine-tuning data sizes

Data Sizes	ULB-MLG[32]	Ninapro DB4[30]
FT-5%	88.50	85.18
FT-20%	89.40	86.15
FT-40%	90.72	87.30
FT-80%	91.00	88.05

Notably, the performance of AEMG-Large approaches or even surpasses the Intra-subject baseline on certain datasets. This result validates our proposed EMG-as-Language paradigm and the effectiveness of the NCT in capturing generalizable physiological features. The performance of existing SOTA EMG methods is arguably limited because they attempt alignment in the feature space, while their input features are still extracted based on fixed-size, fixed-stride windows [14, 33]. This approach disrupts the intrinsic semantic integrity of muscle contraction events, leading to inherent ambiguities in the feature space and thus constraining the effectiveness of domain alignment.

In contrast, the AEMG framework segments the continuous signal into physiologically meaningful *individual muscle contraction tokens* via the NCT. Subsequently, morphological features are learned through VQ reconstruction (Section 3.4), and synergistic features are learned via masked modeling pre-training (Section 3.5). This pipeline allows AEMG to successfully learn a universal EMG vocabulary and motor grammar that span across subjects' physiological differences. Consequently, AEMG extracts more robust and generalizable representations.

The results in Table 2 demonstrate the substantial practical value of the AEMG framework. Using merely 5% of the target user's data, AEMG-Large achieves approximately 90% of the full-data fine-tuning performance on most datasets. This indicates that fine-tuning essentially adapts the pre-trained model—already proficient in general myoelectric language—to acquire user-specific behavioral patterns rather than learning from scratch.

4.2. Evaluation of the Signal Slicing Viewpoint

To further validate the proposed EMG-as-language viewpoint, particularly the effectiveness of NCT (Section 3.2), our experiment compares it against the traditional signal slicing viewpoint, the latter of which employs fixed-size and fixed-stride windows.

For the experimental group using the traditional viewpoint, EMG sentences are created in the same manner as AEMG, but the energy-threshold-based muscle contraction detection process from NCT is completely removed. Instead, the raw signals are blindly segmented into patches of a fixed length. Subsequently, the model under this viewpoint is subjected to VQ reconstruction training (Section 3.4), masked modeling pre-training (Section 3.5), and downstream task evaluation using identical configurations.

During the VQ reconstruction training phase, the proposed NCT viewpoint demonstrates a significant advantage, achieving lower reconstruction loss. This is because the muscle contraction tokens generated by NCT carry clear physiological semantics, *i.e.*, a complete muscle contraction event, which greatly benefits the training of a high-quality, meaningful EMG vocabulary. Conversely, the slices from the traditional viewpoint mix effective contraction segments, static noise, and transitional signals between actions, which increases the ambiguity of vocabulary learning. However, during the masked modeling pre-training phase, an interesting phenomenon is observed. The traditional viewpoint shows higher mask prediction accuracy. This is considered to be primarily attributable to the traditional viewpoint generating a large number of zero-padded tokens or easily predictable static noise tokens, which sig-

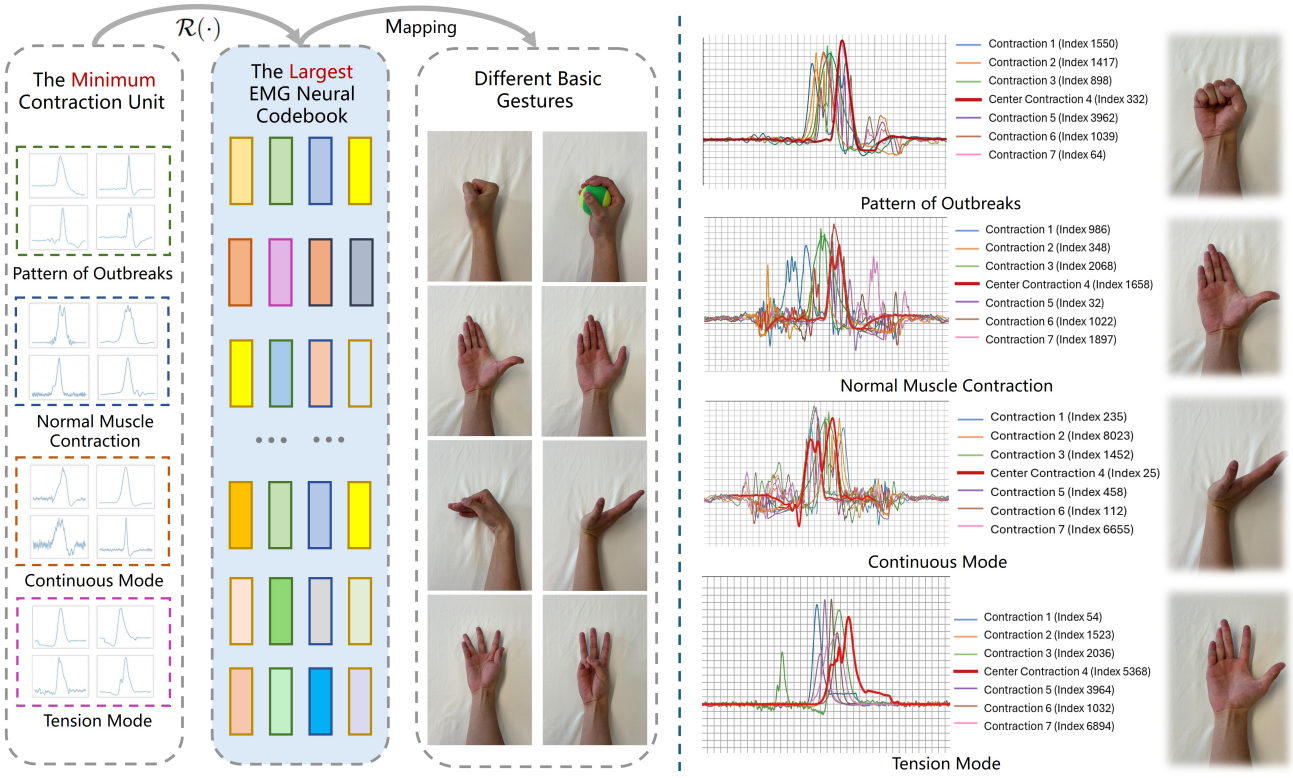


Figure 3. **Illustration of the Neural EMG Vocabulary**. Due to space constraints, only a representative subset is displayed here. The comprehensive gesture modes and corresponding illustrations are detailed in the Appendix. **Left:** Morphological distinctiveness of the EMG vocabulary. Distinct muscle contraction morphologies reliably convey different semantic expressions. **Right:** Context-dependent semantic polysemy in EMG signals. Similar muscle contraction morphologies can convey different functional semantics depending on their spatiotemporal context.

nificantly reduces the difficulty of the pre-training task.

Table 3 shows the performance of these two slicing viewpoints on the downstream gesture classification (LOSO) task. Although the NCT viewpoint has lower pre-training prediction accuracy, it realizes a significant improvement in average accuracy compared to the traditional slicing.

This result strongly confirms the core hypothesis of this work, that the lower prediction accuracy during pre-training precisely reflects the higher modeling difficulty of the pre-training task under the NCT viewpoint. NCT forces the model to understand the real, complex temporal semantics and muscle synergy patterns within EMG sentences, rather than merely reconstructing or predicting simple noise or padded segments. It is this more challenging self-supervised task that enables the model to learn more robust and generalizable representations.

Notably, even when using the suboptimal traditional slicing viewpoint, the average performance of the AEMG framework (82.15%) still surpasses or performs comparably to most SOTA methods shown in Table 1. This strongly demonstrates the inherent advantages of the AEMG's overall architecture in learning EMG representations.

4.3. Illustration of the Neural EMG Vocabulary

To elucidate the mapping between continuous neuromuscular activations and our discrete latent space, we visualize the learned AEMG codebook. Optimized across highly heterogeneous datasets (Section 3.4), the network converges on a universal vocabulary of 8,192 canonical motor primitives.

Morphological analysis of raw signals mapped to identical codebook indices reveals strict physiological clustering (Figure 3). For example, Index 332 exclusively captures transient, high-amplitude bursts indicative of explosive flexor recruitment. Conversely, Index 1452 encodes the sustained, low-amplitude firing typical of continuous extensor tension. This discrete categorization validates the vocabulary's precision in isolating structurally distinct neuromuscular primitives.

Crucially, the latent representation extends beyond isolated waveform morphology to capture sequential motor syntax. We examine raw contraction segments assigned to three distinct codebook indices (348, 1658, and 2068) that exhibit nearly identical localized waveform shapes. Despite their morphological equivalence, the NST dynamically assigns these signals to divergent discrete tokens by integrat-

Table 3. Comparison of accuracy (%) on the gesture classification (LOSO) task between two signal segmentation perspectives.Improvement values are shown in bold green.

Segmentation Perspective	ULB-MLG [32]	EMG-EPN [3]	Ninapro DB4 [30]	DToro-Ossaba [35]	Average
Fixed-size and Fixed-step Time Windows	80.50	85.10	78.00	85.00	82.15
NCT (Ours)	91.50	95.21	88.10	90.30	91.28
Improvement	**+11.00**	**+10.11**	**+10.10**	**+5.30**	**+9.13**

ing the broader spatiotemporal and positional context (Section 3.3). This context-aware mapping demonstrates that the framework successfully resolves the functional role of a contraction within complex, multi-stage gestures, analogous to syntactic disambiguation.

This observation reveals that collective EMG tokens encapsulate morphological, spatiotemporal, and positional features, thereby providing semantically richer embeddings. This is analogous to polysemy in linguistics:

- A given morpheme (similar contraction morphology)
- Derives its meaning from context (spatiotemporal and positional embeddings, *i.e.*, synergistic multi-muscle patterns or sequence position)
- To convey distinct semantics (via different token indices).

For example, a similar activation of the index finger flexor, when contextualized by synergistic thumb activation, may be encoded as part of a thumb flexion (Index 1658). Conversely, when accompanied by co-activation of the other four fingers, it may represent a pinch gesture (Index 348). Most prior EMG representation learning efforts have neglected such spatiotemporal context, limiting their functional semantic richness [15, 34]. The capacity of AEMG to modulate functional semantics through context demonstrates that its learned representations are semantically richer than those of conventional EMG analysis methods.

4.4. Ablation Study: Deconstructing the Key Components of AEMG

In this section, a comprehensive ablation study of the AEMG framework is conducted to quantify the effectiveness of each key component.

w/o EMG Vocabulary: In this experiment, the EMG vocabulary and the VQ process defined in Section 3.4 are removed. Instead, during the masked pre-training stage (Section 3.5), the model is trained to directly reconstruct the masked raw *individual muscle contraction tokens* using a mean squared error (MSE) loss, rather than predicting *collective token* indices. As shown in Table 4, this variant experiences a catastrophic performance drop, declining by 28.3% on ULB-MLG and 25.5% on Ninapro DB4 . This result strongly proves that the vocabulary and VQ discretization are core elements of AEMG's success.

w/o Pre-training: This experiment skips the masked modeling pre-training stage (Section 3.5). The NST encoder is trained using only the VQ reconstruction loss from Section 3.4, which is intended for learning morphology features. It is then directly fine-tuned on downstream tasks. The marked performance decline, exemplified by a 10.2% drop on ULB-MLG, reveals that while VQ reconstruction successfully isolates the neuromuscular vocabulary, it fails to decode the broader movement grammar. Consequently, masked modeling proves indispensable for capturing the synergistic and spatial dependencies inherent in multi-channel EMG sequences

Table 4. Ablation study results for key components of AEMG

Method	ULB-MLG [32]	Ninapro DB4[30]
AEMG-Large (Ours)	**91.50**	**88.10**
w/o EMG Vocabulary	62.3	62.6
w/o Pre-training	81.3	79.5

5. Conclusion

Our study represents a comprehensive theoretical exploration aimed at addressing the generalization challenge that has long plagued the field of myoelectric control. The AEMG framework provides a feasible path toward achieving universal and robust myoelectric signal understanding. Through large-scale pre-training on eight heterogeneous datasets, spanning diverse devices, subjects, and conditions, AEMG has the potential to learn a universal myoelectric language that transcends individual differences. In the most challenging LOSO task, the pre-trained AEMG model demonstrates generalization capabilities that far surpass existing state-of-the-art baseline models. More importantly, in few-shot adaptation tasks, AEMG can rapidly achieve a 90% classification level by fine-tuning on as little as 5% of a new subject's data. The AEMG paradigm is poised to liberate the field of EMG research from building countless isolated, specific models, shifting the focus toward improving a unified and continuously evolving foundational model. This marks a significant shift from the era of custom models, which require extensive user calibration, to the era of data-efficient, rapidly deployable foundation models.

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